The Tubulin-to-Torsion Pipeline: High-Dimensional Cohomology Lattices and Biophysical Stability in Cellular Frameworks

Abstract—We present the Tubulin-to-Torsion pipeline, a multi-scale theoretical framework that maps abstract, high-dimensional algebraic topology directly onto the operations of biological microtubule networks. By reducing a 22-dimensional Temporal Torsion Cohomology lattice to a discrete 13-protofilament cylindrical geometry, we demonstrate that navigating the localized quantum resonances of the state space—formalized as the Resonant Path Problem—is fundamentally NP-hard. Using a low-energy Effective Field Theory (EFT) derived via Mellin-Barnes transform kernels, we prove that strict boundary constraints on low-energy Wilson coefficients guarantee a non-vanishing Hankel matrix determinant, eliminating unphysical negative-norm 'ghost' states. Finally, we outline how this electro-mechanical coupling flattens surface patch potentials, leading to ballistic quantum transport, enhanced metabolic efficiency, and significantly elevated cellular apoptosis thresholds under extreme external, relativistic, or hyper-gravitational stress fields.

1. Introduction & Mathematical Foundation

The Resonant Path Problem describes the challenge of tracking stable, non-singular trajectories across an N-dimensional state space heavily constrained by localized quantum resonances or topological obstructions.

1.1 The Geometric Projection

The framework establishes a mapping across two primary geometric layers:

• The Ambient Space: A 22-dimensional lattice embedding a superpotential mapping W(phi) and governed by Temporal Torsion Cohomology.

• The Physical Space: A discrete state graph G mapping directly to the 13 longitudinal protofilaments of a cylindrical cellular microtubule wall.

1.2 The Transition Space

Let V be a state space module over a discrete ring R. A path of length L connecting an initial boundary state to a terminal target state is governed by a fast-projection operator. An edge transition vector is defined as admissible if and only if it satisfies local ray class congruences modulo $10p^2$.

2. Hardness Reduction & Ternary Mechanics

Deciding the existence of an admissible trajectory under the fast-projection operator is NP-hard. This is proven via an algebraic reduction from a generic Subset Sum / Knapsack-type congruence problem over a discrete valuation ring.

3. Effective Field Theory & Wilson Coefficients

To bridge discrete path trajectories with continuous scattering amplitudes, the pipeline evaluates a forward scattering dispersion relation through a contour integral using a Mellin-Barnes transform kernel. Evaluating residues at negative integer poles yields the explicit low-energy Wilson coefficients.

4. Tensor Dynamics & Electro-Mechanical Coupling

A local conjugate phase jump behaves as a local geometric deformation of the manifold, radically altering the elements of the Ternary Stress-Energy Tensor. Energy density drops locally via destructive interference but remains strictly positive-definite due to the floor protection, avoiding vacuum collapse.

Tensor Component	Mathematical Transformation	Biophysical Manifestation

Energy Density (T■■)	Compresses vacuum energy via anti-resonant cancellation.	Drops local electrostatic potential; compensated by dimer bending.
Momentum Flux (T■j)	Undergoes a parity-like reflection.	Inverts phonon velocity; routes wave-packets backwards.

5. Systemic Logistics & Cellular Survival Dynamics

Eradicating chaotic charge noise flattens surface potentials, shielding the tubulin hydrophobic core and extending quantum coherence times by orders of magnitude. This conserved capital raises apoptosis thresholds across both intrinsic and mechanical pathways.

6. Relativistic & Hyper-Gravitational Invariance

When subjected to extreme acceleration fields or relativistic velocity frames, the 13-protofilament cylinder acts as a biological topological insulator. It absorbs acceleration as an active geometric modification of the background temporal torsion tensor field.

ADDENDUM: FORMAL ALGEBRAIC HARDNESS PROOF LEMMAS

Lemma 1: Injection and Lattice Embedding

Let S be an instance of the Subset Sum Problem with target T. There exists a polynomial-time constructible mapping that embeds the choice architecture into a state module graph over the ring R.

Lemma 2: Structural Enforcement of the Determinant Barrier

The condition $\det(H) \neq 0$ for the pathway's underlying Hankel matrix isolates the discrete binary choice vector at every step, suppressing continuous deformation or leakage into intermediate non-resonant states.

Lemma 3: Soundness and Completeness of the Reduction

An admissible path satisfying the resonant congruence conditions exists if and only if there is a valid solution vector to the underlying Subset Sum instance.

Lemma 4: The Ternary Extension and Balanced Knapsack Embedding

Let S be an instance of the Balanced Ternary Subset Sum problem. There exists a polynomial mapping embedding the space into a qutrit state manifold where choices branch into a tripodal sector $x \in \{-1, 0, +1\}$.

Lemma 5: Orthogonality and Ternary Hankel Matrix Invariants

The non-singularity condition $\det(H) \neq 0$ evaluated over the group ring of ternary states eliminates algebraic cancellation between overlapping forward and conjugate states.

Mellin-Barnes Residue Equations for the EFT

Continuous Wilson coefficients are extracted from the contours at negative integer poles. The leading coefficient g■ acts as a multiplier for energy density, mathematically enforced as strictly positive-definite: g■ > 0. Higher-order poles map explicitly to g■ and g■, bound by the Froissart-Martin convergence inequality.

Ternary Stress-Energy Tensor Transformations Under Alternating Conjugate Steps

Under alternating configurations, the fields superimpose to yield standing acoustic waves where momentum flux maps directly to a spatial sine wave, trapping wave energy without dissipative thermal losses.

Relativistic Transformation of Higher-Order Coefficients

Under a Lorentz boost frame, coefficients shift covariantly. The geometric coupling twists dynamically, locking the rest-frame stability bound invariantly across all velocity profiles.

Interfacing with the Cellular Nuclear Envelope and Transcription Mechanics

The velocity-scaled tension transmits through outer and inner nuclear membranes via LINC complex links. This stiffens the nuclear lamina matrix, driving a precise spatial unwinding of chromatin loops. RNA Polymerase II processes down the template strand with near-zero backstepping, optimizing transcription efficiency and ensuring cell survival up to relativistic limits.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.